



UNIVERSITY OF COLOMBO, SRI LANKA

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I – UCSC AY2023 [held in October 2023]

SCS 1204 – Discrete Mathematics - I

(Two (2) Hours)

Answer ALL questions

Number of Pages = 4

Number of Questions = 4



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Important Instructions to candidates:

- I. Students should answer in the medium of **English language only** using the answer paper provided.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of the answer paper.
- IV. This paper consists of **4** questions in **4** pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Programmable Calculators / Non-Programmable calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

1.

- (a). Show that compound propositions $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are **not** logically equivalent by using truth tables.

[5 Marks]

- (b). Let p, q, r, s, t and u be propositions. Show that the following argument given in symbolic form is valid:

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow (r \wedge s) \\ \neg r \vee (\neg t \vee u) \\ p \wedge t \\ \hline \end{array}$$

$\therefore u$.

[8 Marks]

- (c). Let $L(x, y)$ be the predicate “ x loves y ,” where the domain for both x and y consists of all students at UCSC. Translate each of the following statements into logical expression using predicates, quantifiers, and logical connectives:

- (i). There is somebody whom everybody loves.
- (ii). Everybody loves somebody.
- (iii). There is somebody whom no one loves.
- (iv). Everyone loves himself or herself.
- (v). There is exactly one person whom everybody loves.
- (vi). There is someone who loves no one besides himself or herself.

[6 Marks]

- (d). Translate each of the following nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

- (i). $\forall x \forall y (((x < 0) \wedge (y < 0)) \rightarrow (xy > 0))$
- (ii). $\forall x \forall y \exists z (x + y = z)$
- (iii). $\exists x \exists y ((x^2 > y) \wedge (x < y))$
- (iv). $\exists x \forall y (xy = y)$.

[6 Marks]

2. Prove or disprove each of the following statements:

- (a). For an integer n , if n^2 is divisible by 3, then n is also divisible by 3. [4 Marks]
- (b). For all $n \in \mathbb{N}$, $11^{n+2} + 12^{2n+1}$ is divisible by 133. [5 Marks]
- (c). For any sets A, B, C and D , if $A \times B \subseteq C \times D$ then $A \subseteq C$ and $C \subseteq D$. [4 Marks]
- (d). If p is a prime number greater than 3, then $p^2 = 12k + 1$ for some integer k . [4 Marks]
- (e). If x is a real number such that $\frac{x^2-1}{x+2} > 0$, then either $x > 1$ or $-2 < x < -1$. [4 Marks]
- (f). There exists an integer k such that $4k + 3$ is a perfect square. [4 Marks]

3.

(a). For any $n \in \mathbb{Z}^+$, let $A_n = [-2n, 3n] = \{x \in \mathbb{R} \mid -2n \leq x \leq 3n\}$. Determine each of the following:

- (i). A_2
- (ii). $A_2 - A_3$
- (iii). $A_3 - A_2$
- (iv). $A_2 \triangle A_3$
- (v). $\bigcup_{n \in \mathbb{Z}^+} A_n$
- (vi). $\bigcap_{n \in \mathbb{Z}^+} A_n$.

[6 Marks]

(b). Let A, B and C be three sets. Prove or disprove each of the following statements:

- (i). If $A \cap C = B \cap C$, then $A = B$.
- (ii). If $A \cup C = B \cup C$, then $A = B$.
- (iii). If $(A \cap C = B \cap C)$ and $(A \cup C = B \cup C)$, then $A = B$.
- (iv). If $A \triangle C = B \triangle C$, then $A = B$.

[12 Marks]

(c). Let f be a function from A to B . Let S and T be subsets of B . The subset $f^{-1}(S)$ of A is defined as $f^{-1}(S) = \{x \in A \mid f(x) \in S\}$. Prove each of the following identities:

- (i). $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$
- (ii). $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$
- (iii). $f^{-1}(B - S) = A - f^{-1}(S)$.

[7 Marks]

4.

(a). Let R be a binary relation on $\{1, 2, 3, 4\}$ given by $R = \{(1,2), (1,4), (3,3), (4,1)\}$. Find the smallest relation containing the relation R which is

- (i). reflexive,
- (ii). reflexive and transitive,
- (iii). symmetric and transitive, and
- (iv). reflexive, symmetric and transitive.

[8 Marks]

(b). Let R be a binary relation on a set A .

Let $S = \{(a, b) \mid (a, c) \in R \text{ and } (c, b) \in R \text{ for some } c \in A\}$. Show that if R is an equivalence relation, then S is also an equivalence relation.

[9 Marks]

(c). Suppose that $A = \{3, 5, 9, 15, 24, 45, 60, 72\}$ and R is the partial order relation defined on A by $(x, y) \in R$ if and only if $x \mid y$ (x divides y).

- (i). Draw the Hasse diagram for the relation R .
- (ii). Find all minimal elements of A .
- (iii). Find all maximal elements of A .
- (iv). Find the least element of A (if exists).

[8 Marks]
